`Severity` Class Cheat Sheet

m `Severity(sev_name, exp_attachment=None, exp_limit=inf, sev_mean=0, sev_cv=0, sev_a=nan, sev_b=0, sev_loc=0, sev_scale=0, sev_xs=None, sev_ps=None, sev_wt=1, sev_lb=0, sev_ub=inf, sev_conditional=True, sev_signed=False, sev_reflect=False, name='', note='', hints='', tags=(), doc='', label=None, label_map=None)`

A `Severity` wraps a frozen `scipy.stats` rv (in `fz`) for a single ground-up loss, optionally cut to a layer by the attachment / limit clause. Mixtures and the `wt` weight are handled by `Aggregate`. The following tables show all m methods, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

`name` *object identity*; `label` *display*
`sev_name` *scipy dist name, or* `(c|d)histogram`
`sev_kind`, `long_name` *resolved labels*
`sev_mean`, `sev_cv` *ground-up mean / CV, or*
`sev_a`, `sev_b` *shape parameters 1, 2 (as declared)*
`a`, `b` *resolved shapes handed to scipy*
`sev_loc`, `sev_scale` *shift and scale*
`sev_xs`, `sev_ps` *(d)histogram outcomes / probs*
`sev_wt` *mixture weight (used by* `Aggregate`*)*
`sev_reflect` *reflect about 0*
`program`, `pprogram` *canonical DecL*

2. Update

`exp_attachment`, `exp_limit` *layer inputs*
`attachment`, `limit`, `detachment` *resolved layer*
`sev_lb`, `sev_ub` *conditional support bounds*
`conditional` *losses conditional on attaching the layer (default) or unconditional*
`signed` *negative outcomes allowed*
`pattach`, `pdetach` Pr*(attach),* Pr*(detach)*
`moment_pattach` *attach probability used in the moment integrals*

3. Moments

`sev1`, `sev2`, `sev3` (first three raw moments of the layered severity), `actual_m`, `actual_cv`, `actual_sd`, `actual_var`, `actual_skew` (analytic, post-layer, post-splice), m `moms` (m, cv, skew), m `mean`, m `median`, m `std`, m `var`, m `stats`, m `moment`, m `generic_moment`, m `moment_type`, m `support`

4. Statistical functions

m `cdf`, m `sf`, m `pdf`, m `logcdf`, m `logsf`, m `logpdf`, m `ppf`, m `isf`, m `rvs`, m `interval`, m `expect`, m `entropy`, m `vecentropy`, m `nnlf`

5. Validation

`bounded` (post-layer, post-splice support is bounded), `tail_class`, `tail_description`, `tail_explanation`
No moment / FFT validation: a `Severity` is exact, nothing is discretized until `Aggregate` builds the grid.

6. Output dataframes

`info` (fixed-layout text summary)
No dataframes: there is no grid to tabulate. Discretized severity output lives on `Aggregate` (`sev_density_df`).

7. Reinsurance

None (layer cut only; see Update)

8. Visualization

m `plot`

9. Risk and pricing

None

10. Approximations

m `fit`, m `fit_loc_scale`, m `freeze` (inherited from the scipy rv)

11. Meta

`fz` (the frozen scipy rv), `shapes`, `numargs` (scipy shape info), m `help`, m `format_program`, `pprogram_html`, `note`, `tags`, `doc`, `hints` (the DecL trailer), `label_map`, `labels`, `renamer`, `use_labels`, `random_state`, `xtol`, `badvalue`

Notes:

A `Severity` models one ground-up severity curve; frequency, mixing, and reinsurance live on `Aggregate`. `sev_mean+sev_cv` or `sev_a(+sev_b)` parameterise the shape; `sev_loc/sev_scale` then shift/scale.

The layered severity is $X(a,d) = \min(d, (X-a)^+)$ with attachment a and limit d; `sev1`, `sev2`, `sev3` are its first three raw moments and `actual_*` the derived summary statistics.

Abbreviations: m=mean, cv=coefficient of variation, sd=standard deviation, var=variance, skew(ness).